

High-Efficiency PCSK-CFFH System Design for IoT Networks

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Abstract—Industrial Internet of Things (IIoT) brings opportunities for innovation and upgrading to industrial production, but meanwhile, it poses unprecedented challenges on the physical layer of wireless communications. In this article, we propose a novel frequency hopping transmission scheme for IIoT, which owns much high efficiency and strong anti-jamming ability. This new scheme is built based on the combination of coherent fast frequency hopping (CFFH) and cyclic code shift key (CCSK), named phase cyclic shift-keying-based CFFH (PCSK-CFFH). For the proposed PCSK-CFFH system, the shift of orthogonal base sequence is adopted to modulate the phases among interhop, instead of transmitting each hop with the same phase as conventional CFFH. This scheme is able to further exploit the extra dimensions of CFFH signals and enables the CFFH system to transmit additional information at no extra cost on either bandwidth or power. Moreover, the bit error rate performance of the proposed PCSK-CFFH system is analyzed theoretically. To improve its ability against severe narrow-band jamming, the selection combining is introduced in our scheme. Furthermore, the anti-jamming performance of the proposed PCSK-CFFH system is also demonstrated. Finally, the simulation results are shown to demonstrate that: by introducing the interhop dimension to CFFH, PCSK-CFFH improves the power efficiency and spectrum efficiency by multiple times and simultaneously inherits the excellent anti-jamming performance of the conventional CFFH system. It can be concluded that the proposed PCSK-CFFH scheme has a great potential to be applied to low-power and highly reliable IIoT communications.

Index Terms—Anti-jamming, cyclic code shift key (CCSK), fast frequency hopping (FFH), Industrial Internet of Things (IIoT).

I. INTRODUCTION

WITH the development of Industry 4.0, the Industrial Internet of Things (IIoT) [1], which incorporates cloud computing, big data, embedded computing, and low-power sensors, is being widely used in advanced production processes. IIoT empowers ubiquitous networking and computing capabilities of sensors, machines, and people to enhance the productivity and intelligence of industrial systems. Since it is deployed in critical industrial production sectors, IIoT provides much higher requirements on various issues, e.g.,

security, reliability, and robustness compared to the traditional IoT [2], [3]. The breakdown of IIoT will result in serious safety incidents and economic losses. However, due to the increasing densification of various nodes and the openness of wireless networks, the physical layer of IIoT faces both co-channel interference [4], [5] within internal devices and malicious jamming from external attackers [2], [6]. It is necessary to find a suitable physical-layer anti-jamming scheme for IIoT.

Traditional IoT mainly resorts to spread spectrum (SS) techniques for anti-jamming, such as direct SS (DSS) [5], chirp SS (CSS) [7], frequency hopping SS (FHSS) [8], etc. Compared to other techniques, FHSS is more suitable for IIoT with massive nodes, because it only needs to process narrowband signals after de-hopping and can achieve jamming mitigation at a lower cost. FHSS in IoT is typically slow frequency hopping (SFH) and adopts a “dodge” strategy to mitigate the impact of jamming, e.g., adaptive frequency hopping (AFH) in Bluetooth [9]. Inevitably, this kind of strategy requires a real-time estimation of the state of each channel and changing the hopping dynamically. Hence, it is difficult to cope with the fast-changing interference and jamming in IIoT [9]. Different from SFH, fast frequency hopping (FFH) [10] intrinsically possesses the ability of frequency diversity by operating the carrier hops more than once within a symbol duration. This means that FFH can achieve jamming mitigation through symbol-level diversity combining, rather than dodging jamming. Thus, FFH has an inherent advantage over SFH in terms of anti-jamming ability [11].

In FFH, a noncoherent approach has been analyzed in [12] and [13], namely, fast frequency hopped M-ary orthogonal frequency-shift keying (FFH/MFSK) and achieved wide application. Nevertheless, FFH/MFSK employs noncoherent combining on hops level at the receiver, resulting in a low receiver sensitivity. The larger the frequency-domain diversity gain, the worse the receiver sensitivity performance. In the 1980s, Cherubini and Milstein proposed the coherent FFH (CFFH) system in [14]. They further analyzed the equalization and anti-jamming performance of the CFFH system in [15] and [16]. However, limited by the phase-controlled frequency synthesizer and corresponding synchronization technology at the time, CFFH does not work. Fortunately, in recent years, thanks to the integration of direct digital synthesis (DDS) chips [17]–[20] and the in-depth study of coherent synchronization algorithms [21]–[23], CFFH is re-emerging as trend, and there have been some implementations in the laboratory environments [24], [25].

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IIoT is based on power-limited wireless sensor networks (WSNs) and the efficiency of the transmission has a significant impact on the lifetime of nodes [26]. The transmission efficiency of an IIoT system is directly dependent on the choice of the baseband modulation technique. Previous studies on the CFFH system mostly adopted the phase shift-keying (PSK) scheme, in which hops were modulated with the same phase based on the current symbol, such as BPSK-CFFH [14], [23], [27]. However, the multidimensional character of CFFH on hops implies that this system has the potential to improve efficiency. With this motivation, we resort to multidimensional orthogonal modulation to further exploit the modulation capability of CFFH. The cyclic code shift key (CCSK) [28] is a form of orthogonal M-ary signaling, originally used in tamed SS (TSS). By choosing a base sequence and modulating the carrier with a cyclically shifted version of the sequence, CCSK can achieve multiple times the transmission efficiency of DSS.

In this article, we introduce CCSK to the CFFH system and propose a novel scheme of biorthogonal modulation, named phase cyclic shift-keying-based coherent FFH (PCSK-CFFH). Instead of operating CCSK on a sequence in the time domain, the proposed PCSK-CFFH employs CCSK among the phases of interhop. In other words, the different cyclic shifts of the orthogonal base sequence are modulated on the interhop phases of different symbols and constitute a multidimensional orthogonal space. By embedding additional information in this extra space, each symbol of the proposed PCSK-CFFH can carry more information than the traditional BPSK-CFFH. From the transmission point of view, PCSK-CFFH will bring an inherent coding gain to the system due to the addition of this extra orthogonal space. As a result, the spectrum efficiency and power efficiency of the system are significantly improved. The contributions of this article are summarized as follows.

- 1) In this article, a novel scheme of CFFH is proposed for IIoT, namely, PCSK-CFFH. The proposed scheme introduces the interhop phase dimension as a symbol space to the frequency hopping system and combines the high-efficiency CCSK technology to improve transmission efficiency. We present the PCSK-CFFH system model and the transceiver structure. The complexity of the proposed PCSK-CFFH system relative to the traditional BPSK-CFFH system is also analyzed in-depth, which demonstrates that the proposed PCSK-CFFH scheme is applicable in real scenarios.
- 2) Over AWGN channels, the bit error rate (BER) performance of the proposed PCSK-CFFH system is analyzed. In PCSK-CFFH, the information bits are mapped to the cyclic shift of the base sequence among interhop phases and the symbol-level baseband signals. The total BER and individual BER of those two parts versus E_b/N_0 are derived. Besides, the power efficiency and spectrum efficiency of the PCSK-CFFH system and the BPSK-CFFH system with different spreading gains are compared. The results show that the proposed PCSK-CFFH system is much more efficient than the traditional BPSK-CFFH system.
- 3) In the presence of severe narrow-band jamming, the anti-jamming performance of the proposed PCSK-CFFH

system is analyzed with the selection combining (SC) scheme. For the PCSK-CFFH system, the performance deterioration caused by the nonorthogonality using SC is analyzed analytically. The numerical results are also given for different jamming bandwidths and modulation dimensions, which demonstrates that the PCSK-CFFH system with much higher efficiency has comparable anti-jamming performance to the BPSK-CFFH system.

The remainder of this article is organized as follows. In Section II, we introduce the concept of PCSK-CFFH and describe the transmitter and receiver structure of this system. Subsequently, in Section III, we discuss the BER performance of the PCSK-CFFH system in the presence and absence of jamming. In Section IV, we provide simulation results of PCSK-CFFH against conventional BPSK-CFFH. Finally, Section V concludes this article.

Notations: We use bold uppercase and lowercase letters to represent matrices and vectors, respectively. $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote the transpose, conjugate transpose (Hermitian), and inverse, respectively. We use $\mathbb{E}\{\cdot\}$ to denote expectation, $\Re[\cdot]$ to denote the real part. $\lfloor x \rfloor$ means the rounding down of x and symbol $\odot$ stands for the Hadamard product. $\text{diag}(\mathbf{x})$ represents the diagonal matrix with the diagonal specified by vector $\mathbf{x}$. Function $\text{circshift}(\mathbf{x}, N)$ circularly shifts the elements in $\mathbf{x}$ by N positions and $\delta(t)$ denotes the impulse function. Furthermore, we use $[\mathbf{x}]_k$ to denote the k th element of vector $\mathbf{x}$.

II. PROPOSED PCSK-CFFH SYSTEM

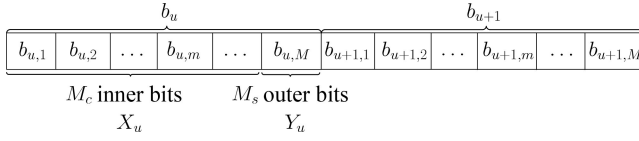
A. PCSK-CFFH Transmitter Design

In a traditional CFFH system [27] with K available frequency channels, there will exist a set of available carrier frequencies, which is denoted as $\mathcal{M} = \{f_1, f_2, \dots, f_k, \dots, f_K\}$. In order to achieve the maximum frequency-domain diversity, each symbol shall traverse all K frequencies of the set $\mathcal{M}$ and occur once at each frequency. For PCSK-CFFH communications, at the transmitter, information transmission is performed symbol by symbol and each symbol has M bits. For the u th symbol, the frequency hopping pattern α_u is a random arrangement of elements in $\mathcal{M}$, which is represented as

$$\alpha_u = [f_{u,1}, f_{u,2}, \dots, f_{u,k}, \dots, f_{u,K}]^T \quad (1)$$

where $f_{u,k}$ denotes the k th hopping frequency. The phase of each hop needs to be regulated in the CFFH system in order to realize coherent receiving. The phase-controlled frequency synthesizer can be achieved by the DDS technology, and we assume that the phase reset to 0 in the frequency synthesizer each time the carrier hops to a new frequency.

In the proposed PCSK-CFFH communication system, the novel idea is to exploit CCSK to modulate the interhop phases. The input binary data is mapped into two parts, i.e., the cyclic shift of the base sequence among interhop phases and the symbol-level baseband signals. As shown in Fig. 1, the input binary data stream is divided into blocks of length M . Each M -bits block is further parsed into M_c inner bits and M_s outer bits, and $M = M_c + M_s$. The number of inner bits M_c satisfies $M_c = \log_2 K$. It is because when there are K available carrier frequencies, based on the cyclic shift of the hop phases

Fig. 1. u th block of information data.

there are K possible hop phase arrangements. As a result, the K possible hop phase arrangements can be mapped to the M_c inner bits. For notational simplicity, in our work, K is the integer power of 2. The outer bits are the information modulated to the baseband signals in the PCSK-CFFH system. Without loss of generality, in the following, we will consider the BPSK modulation scheme, for which $M_s = 1$. Based on the above discussions, the dimensionality of the PCSK-CFFH signal space is $2^M = 2K$. The u th symbol of PCSK-CFFH is $b_u \in [0, 2K - 1]$, which is computed based on the M bits in one symbol such as

$$b_u = \sum_{m=1}^M b_{u,m} 2^{m-1} \quad (2)$$

where $b_{u,m} = 0$ or 1 . The symbols of the inner bits and outer bits are denoted by integers $X_u \in [1, K]$ and $Y_u \in \{-1, 1\}$, respectively, which are

$$\begin{aligned} X_u &= 1 + \sum_{m=1}^{M_c} b_{u,m} 2^{m-1} = 1 + \text{mod}(b_u, 2^{M-1}) \\ Y_u &= 2b_{u,M} - 1 = 2 \left\lfloor \frac{b_u}{2^{M-1}} \right\rfloor - 1. \end{aligned} \quad (3)$$

In the remaining, the signal model of the proposed PCSK-CFFH system will be derived. The transmitter block diagram of the proposed PCSK-CFFH scheme is illustrated in Fig. 2. The base sequence of the u th symbol is

$$\mathbf{c}_u = [c_{u,1}, c_{u,2}, \dots, c_{u,K}]^T \quad (4)$$

where $c_{u,k} \in \{+1, -1\}$ and $\mathbf{c}_u$ has a good autocorrelation performance. After cyclically shifting $\mathbf{c}_u$ by X_u , i.e., $\mathbf{c}_u^{(X_u)} = \text{circshift}(\mathbf{c}_u, X_u \bmod K)$, we have

$$\mathbf{c}_u^{(X_u)} = [c_{u,1}^{(X_u)}, c_{u,2}^{(X_u)}, \dots, c_{u,K}^{(X_u)}]^T. \quad (5)$$

Then at the k th hop, the baseband signal can be represented by

$$S_{u,k} = Y_u c_{u,k}^{(X_u)}. \quad (6)$$

Together with (1), the transmit signal with normalized power at each hop is

$$\begin{aligned} S_{\text{PCSK-CFFH}}(t) &= \sum_{u=1}^U \sum_{k=1}^K Y_u c_{u,k}^{(X_u)} \cos(2\pi f_{u,k} t) \\ &\times g_{T_h}(t - kT_h) * \delta(t - uT_s) \end{aligned} \quad (7)$$

where T_h and T_s are the hop duration and symbol duration, respectively, and $T_s = KT_h$. The function $g_{T_h}(t)$ is a gate function of width T_h , i.e.,

$$g_{T_h}(t) = \begin{cases} 1, & 0 \leq t < 1 \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Based on the signal model in (7), it is worth noting that when all elements of $\mathbf{c}_u^{(X_u)}$ are equal to 1, the proposed PCSK-CFFH system will degenerate to the traditional BPSK-CFFH system. It is obvious that BPSK-CFFH is a special case of PCSK-CFFH. Different from the traditional BPSK-CFFH system, the proposed PCSK-CFFH system has K -extra dimensions representing the inner bits of X_u . The bandwidth of each hop is set to be the main lobe width of $g_{T_h}(t)$, which is $(2/T_h)$. Define the total CFFH bandwidth B as the frequency interval from the lower edge of the f_1 channel to the upper edge of the f_K channel. B is expressed as

$$B = (K - 1)\Delta f + \frac{2}{T_h} \quad (9)$$

where Δf denotes the adjacent frequency channel interval. To maintain the carrier phase difference constant during channel switching, Δf should be chosen so that $\Delta f T_h$ is an integer. Based on the definitions of X_u and Y_u in (3), it is obvious that X_u and Y_u are independent of each other in (7). Based on the bandwidth given in (9) and together with the fact that in each symbol there are M bits to transmit, the spectrum efficiency of the PCSK-CFFH system is

$$\gamma_{\text{PCSK-CFFH}} = \frac{M}{B} = \frac{1 + \log_2 K}{(K - 1)\Delta f T_s + 2K} \quad (10)$$

where the second equality is based on the equation $M = M_c + M_s = \log_2 K + 1$. On the other hand, BPSK-CFFH transmits 1 bit of information per symbol within the same bandwidth, so the spectrum efficiency of the BPSK-CFFH system is

$$\gamma_{\text{BPSK-CFFH}} = \frac{1}{(K - 1)\Delta f T_s + 2K}. \quad (11)$$

It is obvious that the spectrum efficiency of the PCSK-CFFH system is M times that of the BPSK-CFFH system.

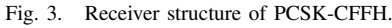
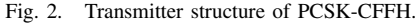
B. PCSK-CFFH Receiver Design

In this section, the full digital receiver of PCSK-CFFH is discussed. As mentioned in Section II-A, the PCSK-CFFH signal can degenerate to the BPSK-CFFH signal via changing all elements of $\mathbf{c}_u^{(X_u)}$ in (7) are equal to 1. As this operation is independent of synchronization, the pilot-aided synchronization scheme used in the traditional CFFH system [21], [23] can also be applied to the PCSK-CFFH system. The remaining key work for the PCSK-CFFH system design is the demodulation. To facilitate analysis, in the following derivations and discussions, it is assumed that the system has been ideally synchronized and the phase scatter among hops due to the delay has been compensated.

The received signal model is given based on the structure of the PCSK-CFFH receiver depicted in Fig. 3. After decimation, the u th symbol is

$$\mathbf{z}_u = \mathbf{r}_u + \mathbf{n}_u + \mathbf{J}_u \quad (12)$$

where $\mathbf{r}_u = [r_{u,1}, r_{u,2}, \dots, r_{u,K}]^T$ denotes the signal sequence of the u th symbol. $\mathbf{n}_u = [n_{u,1}, n_{u,2}, \dots, n_{u,K}]^T$ and $\mathbf{J}_u = [J_1, J_2, \dots, J_K]^T$ denote the sampling sequences of complex noise and jamming after matched filtering and decimation,



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$$\begin{aligned}\hat{b}_u &= \arg \max_{0 \leq b_u \leq 2K-1} \{P(b_u)P(\mathbf{z}_u|b_u)\} \\ &= \arg \max_{0 \leq b_u \leq 2K-1} \{P(b_u)P_{\hat{\mathbf{n}}_u}(\mathbf{z}_u - \mathbf{r}_u)\}.\end{aligned}\quad (21)$$

As each symbol is transmitted with the same probability, $P(b_u)$ can be ignored in (21). By expanding $P_{\hat{\mathbf{n}}_u}(\mathbf{z}_u - \mathbf{r}_u)$ in (21) and based on (19), the ML estimate of the received symbol $\hat{b}_u$ in (21) can be rewritten as

$$\begin{aligned}\hat{b}_u &= \arg \max_{0 \leq b_u \leq 2K-1} \left\{ \frac{1}{\pi^K |\Lambda_{\hat{\mathbf{n}}_u}|} \exp \left[-(\mathbf{z}_u - \mathbf{r}_u)^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} (\mathbf{z}_u - \mathbf{r}_u) \right] \right\} \\ &= \arg \max_{0 \leq b_u \leq 2K-1} \left\{ \left[-(\mathbf{z}_u - \mathbf{r}_u)^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} (\mathbf{z}_u - \mathbf{r}_u) \right] \right\} \\ &= \arg \max_{0 \leq b_u \leq 2K-1} \left\{ 2\Re \left[\mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{r}_u \right] - \mathbf{r}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{r}_u \right\}.\end{aligned}\quad (22)$$

For the flat fading channels, i.e., $\mathbf{A} = \mathbf{I}$, $\mathbf{r}_u$ is rewritten as

$$\mathbf{r}_u = \frac{\sqrt{E_s R_s}}{K} Y_u \mathbf{C}_u \mathbf{p}_{X_u} \quad (23)$$

based on which, $\hat{b}_u$ in (22) further equals

$$\begin{aligned}\hat{b}_u &= \arg \max_{0 \leq b_u \leq 2K-1} \left\{ 2\Re \left[\frac{\sqrt{E_s R_s}}{K} Y_u \mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{X_u} \right] \right. \\ &\quad \left. - \frac{E_s R_s}{K^2} \mathbf{p}_{X_u}^H \mathbf{C}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{X_u} \right\}.\end{aligned}\quad (24)$$

Due to the orthogonality of the columns of $\mathbf{C}_u$, we have

$$\frac{E_s R_s}{K^2} \mathbf{p}_{X_u}^H \mathbf{C}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{X_u} = \frac{E_s R_s}{K^2} \sum_{k=1}^K \frac{1}{\sigma_n^2 + P_{J_k}} \quad (25)$$

which is a constant and can be removed directly. Then, the ML estimate of the received symbol $\hat{b}_u$ in (24) is finally equivalent to

$$\hat{b}_u = \arg \max_{0 \leq b_u \leq 2K-1} \left\{ \Re \left[Y_u \mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{X_u} \right] \right\}.\quad (26)$$

It should be noted that sequence $\mathbf{p}_{X_u}$ has only the X_u th element equal 1 and all columns of $\mathbf{C}_u$ are cyclic shift versions of the vector $\mathbf{c}_u$. Introduce sequence $\mathbf{l}_u$ as

$$[\mathbf{l}_u]_{l_u} = \Re \left[\mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{l_u} \right], \quad l_u \in [1, K]. \quad (27)$$

The optimal decisions of X_u and Y_u are denoted as $\hat{X}_u$ and $\hat{Y}_u$, respectively, and they can be determined by the peak position and peak phase of $\mathbf{l}_u$, i.e.,

$$\begin{aligned}\hat{X}_u &= \text{findpeak}(|\mathbf{l}_u|) \\ \hat{Y}_u &= \text{sgn}([\mathbf{l}_u]_{\hat{X}_u})\end{aligned}\quad (28)$$

where function $\text{findpeak}(\mathbf{x})$ returns the minimum index with the maxima (peaks) of the $\mathbf{x}$ and $\text{sgn}(\cdot)$ is defined as

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0. \end{cases} \quad (29)$$

Based on (28), the ML estimate for the received symbol $\hat{b}_u$ equals

$$\hat{b}_u = 2^{M-2} (\hat{Y}_u + 1) + (\hat{X}_u - 1). \quad (30)$$

In the implementation, the circular correlation can be replaced by discrete Fourier transform (DFT) as in Fig. 3. By this means, $\mathbf{l}_u$ can be given by

$$\mathbf{l}_u = \Re \left[\text{DFT}(\text{DFT}(\mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1})^T \odot \text{conj}(\text{DFT}(\mathbf{c}_u))) \right]. \quad (31)$$

The above analysis shows that the PCSK-CFFH optimal reception claims a known of $\Lambda_{\hat{\mathbf{n}}_u}^{-1}$, which requires the accurate estimation of noise power, jamming power, and signal power of each hop. It obviously increases the complexity of PCSK-CFFH implementation and is inappropriate on the IoT nodes. Despite all this, according to the definition of $\Lambda_{\hat{\mathbf{n}}_u}$, the diagonal elements of $\Lambda_{\hat{\mathbf{n}}_u}^{-1}$ corresponding to the jammed frequency channel would be close to 0 under severe narrow-band jamming conditions, in which case we can still employ SC as the suboptimization solution, i.e., only the jamming-free channels are combined.

In the following, the anti-jamming performance of the PCSK-CFFH system will be analyzed, with this suboptimal solution.

The concept of biorthogonal modulation gives PCSK-CFFH a higher modulation order than BPSK-CFFH. The tradeoff is a rise in resources. Increased resources are the storage resource for $\mathbf{c}_u$ at the transmitter, which is almost negligible. For the receiver, the increased processing resources compared to BPSK-CFFH are mainly lying in the dashed box in Fig. 3. Since K is an integer power of 2, the DFT in the above algorithm can be replaced by the fast Fourier transform (FFT). In addition, the transform and conjugate results of the base sequence $\mathbf{c}_u$ can be stored in advance. Then, the increased computing resources are mainly caused by the FFT module of the input signal, the inverse FFT (IFFT) module, and the peak search process. The overall time complexity of the above process is $\mathcal{O}(K \log_2 K)$. It should be noted that the FFT and IFFT algorithms are especially optimized by many hardware platforms, including MCUs and DSPs. That means PCSK-CFFH can still be implemented even for conditions with large K .

III. PERFORMANCE ANALYSIS OF PCSK-CFFH

In this section, various performance metrics of the proposed PCSK-CFFH system are analyzed. The BER and power efficiency of the proposed PCSK-CFFH system are analyzed first. For completeness, the BERs for the inner bits, outer bits, and total bits are analyzed separately. The anti-jamming performance is also analyzed for the proposed PCSK-CFFH system. Then, a brief comparison between the proposed PCSK-CFFH and traditional BPSK-CFFH systems will be given.

A. BER Performance and Power Efficiency Analysis

In (27), $\mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1} \mathbf{C}_u \mathbf{p}_{l_u}$ is the received sequence $\mathbf{z}_u$ multiplied by the diagonal elements of $\Lambda_{\hat{\mathbf{n}}_u}^{-1}$, which is then cyclically correlated with $\mathbf{c}_u$. Therefore, $\Lambda_{\hat{\mathbf{n}}_u}^{-1}$ has the function to provide diversity factors for each hop input. When there is no jamming ($P_{J_k} = 0$), elements on the diagonal of $\Lambda_{\hat{\mathbf{n}}_u}^{-1}$ are the same and the result of $\mathbf{z}_u^H \Lambda_{\hat{\mathbf{n}}_u}^{-1}$ is equivalent to multiply the input signal and noise by a constant which has no effect on

receiving. Therefore, we can define a combining coefficient matrix $\mathbf{T} = \mathbf{I}$ to replace $\Lambda_{\hat{n}_u}^{-1}$ in (27). For outer bits, $Y_u = +1$ and $Y_u = -1$ correspond to the same error probability. To simplify the analysis, the transmitting Y_u will be assumed to $+1$ in the following. Substituting (12) and (23) into (27), $\mathbf{l}_u$ in (27) becomes

$$[\mathbf{l}_u]_{l_u} = \Re \left[\frac{\sqrt{E_s R_s}}{K} \mathbf{p}_{X_u}^H \mathbf{C}_u^H \mathbf{C}_u \mathbf{p}_{l_u} + \mathbf{n}_u^H \mathbf{C}_u \mathbf{p}_{l_u} \right] \quad (32)$$

where a constant scaling factor $1/\sigma_n^2$ has been ignored with the substitution of $\mathbf{T}$ for $\Lambda_{\hat{n}_u}^{-1}$. Due to the orthogonality of the columns of $\mathbf{C}_u$, when $l_u = X_u$, $[\mathbf{l}_u]_{l_u}$ is

$$\begin{aligned} [\mathbf{l}_u]_{X_u} &= \Re \left[\frac{\sqrt{E_s R_s}}{K} \mathbf{p}_{X_u}^H \mathbf{C}_u^H \mathbf{C}_u \mathbf{p}_{X_u} + \mathbf{n}_u^H \mathbf{C}_u \mathbf{p}_{X_u} \right] \\ &= \underbrace{\sqrt{E_s R_s}}_{\text{Signal}} + \underbrace{\Re[\mathbf{n}_u^H \mathbf{C}_u \mathbf{p}_{X_u}]}_{\text{Noise}} \end{aligned} \quad (33)$$

and otherwise when $l_u \neq X_u$ and by denoting $l_u = \bar{X}_u$, $[\mathbf{l}_u]_{\bar{X}_u}$ is

$$[\mathbf{l}_u]_{\bar{X}_u} = \Re[\mathbf{n}_u^H \mathbf{C}_u \mathbf{p}_{\bar{X}_u}]. \quad (34)$$

The noise component is the real part of the sum of each hop noise multiplied by a constant orthogonal sequence, which is still Gaussian distributed. After combining, the symbol-level noise power is $K\sigma_n^2/2$, and the distribution of $[\mathbf{l}_u]_{l_u}$ is

$$[\mathbf{l}_u]_{l_u} \sim \mathcal{N}\left(\sqrt{E_s R_s}, \frac{K\sigma_n^2}{2}\right), \quad l_u = X_u \quad (35)$$

$$[\mathbf{l}_u]_{l_u} \sim \mathcal{N}\left(0, \frac{K\sigma_n^2}{2}\right), \quad l_u = \bar{X}_u. \quad (36)$$

Recall that the peak phase and peak position of $\mathbf{l}_u$ determine the estimates of inner bits and outer bits. The conditions for the correct decision of b_u are

$$|[\mathbf{l}_u]_{\bar{X}_u}| < |[\mathbf{l}_u]_{X_u}|, \quad \forall \bar{X}_u \quad (37)$$

$$\text{sgn}([\mathbf{l}_u]_{X_u}) = Y_u = +1 \quad (38)$$

based on which and defining

$$\begin{aligned} v &= \sqrt{\frac{2}{N_0 R_s}} [\mathbf{l}_u]_{X_u} - \sqrt{\frac{2E_s}{N_0}} \\ x &= \sqrt{\frac{2}{N_0 R_s}} [\mathbf{l}_u]_{\bar{X}_u} \\ v' &= [\mathbf{l}_u]_{X_u} \text{ and } x' = [\mathbf{l}_u]_{\bar{X}_u} \end{aligned} \quad (39)$$

where $N_0 = (K\sigma_n^2/R_s)$ is the single-sided power spectral density. Based on the distributions in (35) and (36), the probability of correct reception of the symbol can be expressed as

$$\begin{aligned} P_c^{(b_u)}(K, \gamma) &= \int_0^\infty \left[\int_{-v'}^{v'} P(x') dx' \right]^{K-1} P(v') dv' \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{\frac{2E_s}{N_0}}}^\infty \left[\frac{1}{\sqrt{2\pi}} \int_{-v+\sqrt{\frac{2E_s}{N_0}}}^{v+\sqrt{\frac{2E_s}{N_0}}} e^{-\frac{x^2}{2}} dx \right]^{K-1} e^{-\frac{v^2}{2}} dv \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{2M\gamma}}^\infty \left[1 - 2Q(v + \sqrt{2M\gamma}) \right]^{K-1} e^{-\frac{v^2}{2}} dv \end{aligned} \quad (40)$$

where $\gamma = E_b/N_0$ is the bit-level signal-to-noise ratio, and $Q(x) = (1/\sqrt{2\pi}) \int_x^\infty e^{-(x^2/2)} dx$. Then, the symbol error rate (SER) can be presented as

$$P_e^{(b_u)}(K, \gamma) = 1 - P_c^{(b_u)}(K, \gamma). \quad (41)$$

Now, we discuss the BER in the two dimensions of inner bits and outer bits. For inner bits, the condition for a correct decision is given in (37) and therefore the error probability of $\hat{X}_u$ is

$$P_e^{(X_u)}(K, \gamma) = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \left[1 - 2Q(v + \sqrt{2M\gamma}) \right]^{K-1} e^{-\frac{v^2}{2}} dv. \quad (42)$$

Based on (42), the average error of the inner bits will be derived. Let w_n denote the number of cases where there are n errors in $\log_2 K$ inner bits, which equals

$$w_n = \binom{\log_2 K}{n}. \quad (43)$$

Then, the average error bits of inner bits can be expressed as

$$\mathbb{E}\{w_n\} = \frac{w_1 + 2w_2 + 3w_3 + \dots + (\log_2 K)w_{\log_2 K}}{K-1} \quad (44)$$

where $K-1$ is the total number of error cases. Let $N = \log_2 K$, by considering the symmetry of the binomial coefficients, the average error bits of inner bits in (44) can be derived as

$$\begin{aligned} \mathbb{E}\{w_n\} &= \frac{1}{K-1} \sum_{n=0}^{\log_2 K} n w_n \\ &= \begin{cases} \frac{1}{K-1} \left\{ [0w_0 + Nw_N] + [1w_1 + (N-1)w_{N-1}] + \dots \right. \\ \quad \left. + \left[\frac{N+1}{2} w_{\frac{N+1}{2}} + \frac{N-1}{2} w_{\frac{N-1}{2}} \right] \right\}, & N \text{ is odd} \\ \frac{1}{K-1} \left\{ [0w_0 + Nw_N] + [1w_1 + (N-1)w_{N-1}] + \dots \right. \\ \quad \left. + \frac{N}{2} w_{\frac{N}{2}} \right\}, & N \text{ is even} \end{cases} \\ &= \frac{N}{2(K-1)} \sum_{n=0}^N w_n. \end{aligned} \quad (45)$$

According to the binomial theorem, we get $2^N = \sum_{n=0}^N w_n$, leading to

$$\mathbb{E}\{w_n\} = \frac{N}{2(K-1)} 2^N = \frac{\log_2 K}{2(K-1)} 2^{\log_2 K} = \frac{K \log_2 K}{2(K-1)}. \quad (46)$$

The BER of inner bits is thereby shown as

$$\begin{aligned} P_b^{(X_u)}(K, \gamma) &= \frac{\mathbb{E}\{w_n\} P_e^{(X_u)}}{\log_2 K} \\ &= \frac{K}{2(K-1)} P_e^{(X_u)}(K, \gamma). \end{aligned} \quad (47)$$

For the outer bits, the bit errors can be classified into the following two types.

- 1) *Type I Error*: The peak position of $\mathbf{l}_u$ is detected correctly, i.e., $\hat{X}_u = X_u$. The probability of bit error can be calculated based on the BER of coherently detected BPSK at the peak position, given by

$$P_{b1}^{(Y_u)}(K, \gamma) = Q(\sqrt{2M\gamma}). \quad (48)$$

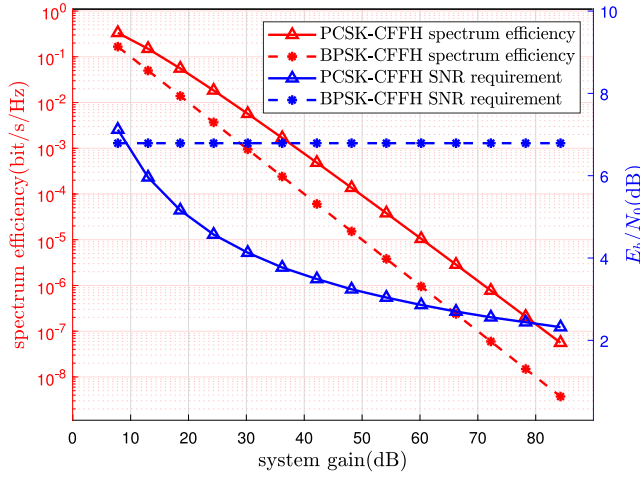


Fig. 4. Comparison of the efficiency of PCSK-CFFH and BPSK-CFFH.

- 2) *Type II Error*: The peak position of $\mathbf{l}_u$ is detected incorrectly, i.e., $\hat{X}_u = \bar{X}_u$. In this case, the estimate $\hat{Y}_u$ entirely depends on the noise, and the probability of bit error is

$$P_{b2}^{(Y_u)} = \frac{1}{2}. \quad (49)$$

Therefore, the BER of outer bits (SER is the same) can be expressed as

$$\begin{aligned} P_b^{(Y_u)}(K, \gamma) &= P_{b1}^{(Y_u)}(1 - P_e^{(X_u)}(K, \gamma)) + P_{b2}^{(Y_u)} P_e^{(X_u)}(K, \gamma) \\ &= Q(\sqrt{2M\gamma})(1 - P_e^{(X_u)}(K, \gamma)) + \frac{1}{2} P_e^{(X_u)}(K, \gamma). \end{aligned} \quad (50)$$

The overall BER of the PCSK-CFFH system can be calculated via the combination of the BERs of the inner bits and outer bits cases, given by (51), shown at the bottom of the page.

To reveal the physical meanings, we consider the case when K is relatively large. Specifically, the bit-level signal-to-noise ratio and K interval of interest are $\gamma \in [0 \text{ dB}, 8 \text{ dB}]$ and $K \geq 512$, respectively. Then, $(K/2(K-1)) \rightarrow (1/2)$ and $Q(\sqrt{2M\gamma}) \leq Q(\sqrt{18}) \ll (1/2)$. As a result, (47), (50), and (51) can be further simplified into the following results:

$$\begin{aligned} P_b^{(X_u)}(K, \gamma) &\approx \frac{1}{2} P_e^{(X_u)}(K, \gamma) \\ P_b^{(Y_u)}(K, \gamma) &\approx Q(\sqrt{2M\gamma}) + \frac{1}{2} P_e^{(X_u)}(K, \gamma) \\ P_b^{(b_u)}(K, \gamma) &\approx \frac{Q(\sqrt{2M\gamma})}{M} + \frac{1}{2} P_e^{(X_u)}(K, \gamma). \end{aligned} \quad (52)$$

By assuming $\Delta f = (1/T_h)$, Fig. 4 gives an intuitive illustration for comparisons between the transmission efficiency of the PCSK-CFFH system and the BPSK-CFFH system for $K = 2^1, 2^2, \dots, 2^{14}$. The spectrum efficiency curves of these two schemes are based on (10) and (11). The power efficiency is based on the value of the SNR requirement to achieve the corresponding BER. For the BER of 10^{-3} , the curves of (51) and ideal BER versus E_b/N_0 result are plotted in Fig. 4. The system gain is defined as a total bandwidth over symbol rate, i.e., $10 \lg(K^2 + K)$. It shows that the PCSK-CFFH system is superior to the BPSK-CFFH system in terms of power efficiency and spectrum efficiency when K is larger than 4. The larger the K , the more pronounced the trend.

B. Anti-Jamming Performance With Selection Combining in PCSK-CFFH

The BER performance analysis of the PCSK-CFFH system in the absence of jamming has been developed above. In this section, we analyze the anti-jamming performance of the system in the presence of narrow-band jamming with the SC scheme. SC directly zeroes the jammed channel and receives only the input of the jamming-free channel. However, removing jammed channels also causes a loss of signal energy, which deteriorates the signal-to-noise ratio. Assuming that the ideal filter is deployed, this loss is characterized by the E_b/N_0 back-off (the additional E_b/N_0 required to achieve the same BER condition as in the absence of jamming) which is demonstrated on the BER curves as the leftward shift of the curve. Let V denote the number of jammed hops, and $\lambda = (V/K)$ be the ratio of jammed bandwidth to total bandwidth. Then, there are V zeros and $K - V$ ones on diagonal of the combining coefficient matrix $\mathbf{T}$. The E_b/N_0 backoff for the BPSK-CFFH system with SC can be expressed as [11]

$$[B_{\text{BPSK-CFFH}}]_{\text{dB}} = 10 \lg(1 - \lambda). \quad (53)$$

For the proposed PCSK-CFFH system, following the idea of the peak detection in the previous section, the distribution of the X_u th element of $\mathbf{l}_u$ can be written as

$$[\mathbf{l}_u]_{X_u} \sim \mathcal{N}\left((1 - \lambda)\sqrt{E_s R_s}, \frac{(1 - \lambda)K\sigma_n^2}{2}\right). \quad (54)$$

Whereas for the element at position $\bar{X}_u$, the mean values are not strictly zeros. Since zeros on the diagonal of the combining matrix $\mathbf{T}$ will zero the corresponding row of $\mathbf{T}\mathbf{C}_u$, the orthogonality of $\mathbf{T}\mathbf{C}_u$ and $\mathbf{C}_u$ is disrupted. As a result, the

$$\begin{aligned} P_b^{(b_u)}(K, \gamma) &= \frac{P_{b1}^{(Y_u)}(K, \gamma)}{M} + \frac{P_{b2}^{(Y_u)}(K, \gamma) + P_b^{(X_u)}(K, \gamma)}{M} \\ &= P_e^{(X_u)}(K, \gamma) \left[\frac{K \log_2 K}{2M(K-1)} + \frac{1}{2M} \right] + \frac{Q(\sqrt{2M\gamma})}{M} [1 - P_e^{(X_u)}(K, \gamma)] \\ &= \frac{Q(\sqrt{2M\gamma})}{M} + P_e^{(X_u)}(K, \gamma) \left[\frac{MK - 1}{2M(K-1)} - \frac{Q(\sqrt{2M\gamma})}{M} \right] \end{aligned} \quad (51)$$

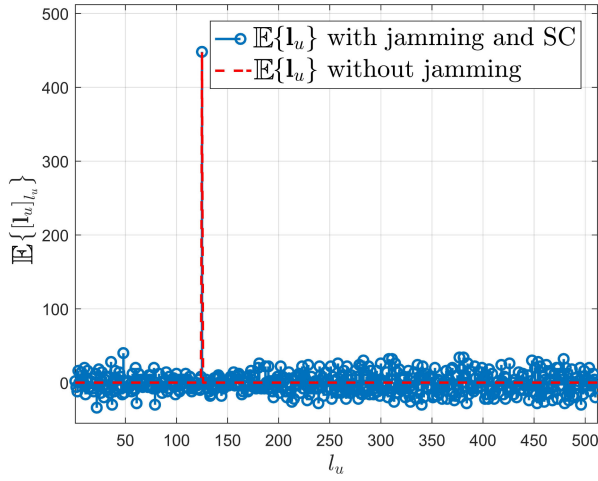


Fig. 5. Mean value of $\mathbf{l}_u$. Nonorthogonal causes “burr” at the position of $\bar{X}_u$. The parameters are $K = 512$ and $V = 64$.

mean value of $[\mathbf{l}_u]_{\bar{X}_u}$ is expressed as

$$\mathbb{E}\{[\mathbf{l}_u]_{\bar{X}_u}\} = \Re \left[\frac{\sqrt{E_s R_s}}{K} \mathbf{p}_{X_u}^H \mathbf{C}_u^H \mathbf{T} \mathbf{C}_u \mathbf{p}_{\bar{X}_u} \right] \quad (55)$$

is no longer constant at 0. As illustrated in Fig. 5, $\mathbb{E}\{\mathbf{l}_u\}$ exhibits a random characteristic.

Obviously, for different jamming, the hops to be zeroed are different and the sidelobes of $\mathbb{E}\{\mathbf{l}_u\}$ are different. The number of subsets deleted exactly V hops is $\binom{K}{V} = (K!/(K-V)!V!)$. Even for moderate values of K and V , this number can be too enormous to analyze BER in analytical forms. For $K = 512$ and $V = 64$, for instance, $\binom{K}{V} \sim 10^{82}$. The direct analysis of the post-zeroing performance of PCSK-CFFH for general cases is very challenging.

To facilitate the analytical analysis, the impact of SC will be analyzed from a probability perspective. As remarked earlier, for the u th symbol, the frequency hopping pattern α_u is a random arrangement of elements in $\mathcal{M}$. For a fixed jamming, since a different α_u is adopted for each symbol, the 0 elements of $\mathbf{l}_u$ are randomly arranged. As the above definition, $\mathbf{c}_u^{(X_u)}$ is the X_u th column of $\mathbf{C}_u$, i.e., $(\mathbf{c}_u^{(X_u)})^H = \mathbf{p}_{X_u}^H \mathbf{C}_u^H$. The deleted component of $\mathbf{c}_u^{(X_u)}$ equals

$$\bar{\mathbf{d}}_u^H = (\mathbf{c}_u^{(X_u)})^H - \mathbf{p}_{X_u}^H \mathbf{C}_u^H \mathbf{T} \quad (56)$$

and the number of 1 in $\bar{\mathbf{d}}_u^H$ is V . Then, we analyze the distribution of the mean of $[\mathbf{l}_u]_{\bar{X}_u}$. Substituting (56) into (55), the mean of $[\mathbf{l}_u]_{\bar{X}_u}$ is

$$\mathbb{E}\{[\mathbf{l}_u]_{\bar{X}_u}\} = -\frac{\sqrt{E_s R_s}}{K} \beta_{u, \bar{X}_u} \quad (57)$$

where $\beta_{u, \bar{X}_u}$ is defined as

$$\beta_{u, \bar{X}_u} = \Re \left[\bar{\mathbf{d}}_u^H \mathbf{c}_u^{(\bar{X}_u)} \right]. \quad (58)$$

Further, let $\kappa_{u, \bar{X}_u} = \mathbf{c}_u^{(X_u)} \odot \mathbf{c}_u^{(\bar{X}_u)}$, the number of +1 and -1 elements in $\kappa_{u, \bar{X}_u}$ must be the same due to the orthogonality of $\mathbf{C}_u$. $\beta_{u, \bar{X}_u}$ can be regarded as the sum of V elements randomly

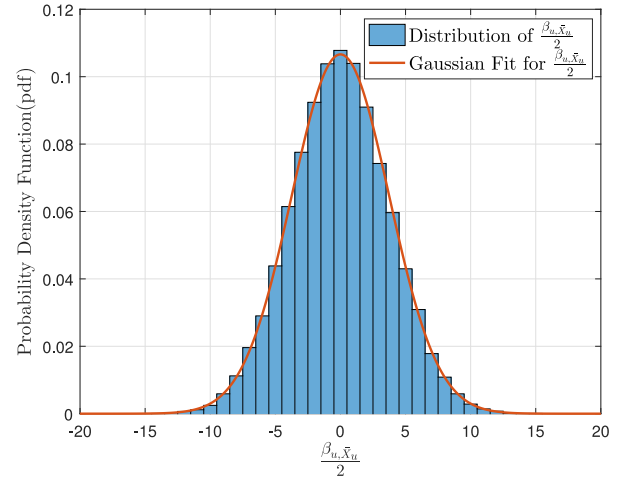


Fig. 6. Histogram for $(\beta_{u, \bar{X}_u})/2$ along an arbitrary $\bar{X}_u$ of $4e^4$ symbols. The parameter are $K = 512$ and $V = 64$, and the corresponding Gaussian fits are overlaid.

selected from $\kappa_{u, \bar{X}_u}$. Defining the number of +1 among these V elements as i , then i satisfies the hypergeometric distribution as

$$i \sim \mathcal{H}\left(V, \frac{K}{2}, K\right). \quad (59)$$

It should be noted that, in general, when K is large enough and λ in (54) is not close to 0 or 1, the above hypergeometric distribution is approximately a binomial distribution. With the De Moivre–Laplace theorem, this binomial distribution can be further approximated by a Gaussian distribution [29], [30], when K is a large number. Let the superscript “ $\sim$ ” denote the approximate distribution, and $\tilde{i} \sim \mathcal{N}(\mu(\tilde{i}), \sigma^2(\tilde{i}))$. It is easy to obtain $\mu(\tilde{i}) = (\lambda K/2)$ and $\sigma^2(\tilde{i}) = (\lambda(1-\lambda)K/4)$. As a result, for $\beta_{u, \bar{X}_u}$ and $\tilde{\beta}_{u, \bar{X}_u}$, we can get

$$\frac{\beta_{u, \bar{X}_u}}{2} = \frac{i - (\lambda K/2)}{2} = i - \frac{\lambda K}{2} \quad (60)$$

and

$$\frac{\tilde{\beta}_{u, \bar{X}_u}}{2} \sim \mathcal{N}\left(0, \frac{\lambda(1-\lambda)K}{4}\right). \quad (61)$$

Fig. 6 depicts the goodness of a Gaussian fit to the empirical distribution of the $(\beta_{u, \bar{X}_u})/2$. It can be seen from this example that the distribution of $(\beta_{u, \bar{X}_u})/2$ is approximately Gaussian distribution when $K \geq 512$.

Now, it can be realized that the nonorthogonal effect caused by the SC is approximately equivalent to a Gaussian distribution. By (57) and (60), we have

$$\mathbb{E}\left\{[\tilde{\mathbf{l}}_u]_{\bar{X}_u}\right\} \sim \mathcal{N}\left(0, \lambda(1-\lambda) \frac{E_s R_s}{K}\right). \quad (62)$$

For each symbol, $[\mathbf{l}_u]_{\bar{X}_u}$ is the superposition of this distribution with noise, which further leads to

$$[\tilde{\mathbf{l}}_u]_{\bar{X}_u} \sim \mathcal{N}\left(0, \lambda(1-\lambda) \frac{E_s R_s}{K} + \frac{(1-\lambda)K\sigma_n^2}{2}\right). \quad (63)$$

Through the approximate distribution of $\mathbf{l}_u$, we can derive the approximate BER performance of the PCSK-CFFH system with SC. Adopting the same method as that in Section III-A

and distributions in (54) and (63), the approximate SER can be derived as (64), shown at the bottom of the page, and approximate BER of inner bits and outer bits as (65)–(67), shown at the bottom of the page. The overall BER $\tilde{P}_e^{(b_u)}(K, \lambda, \gamma)$ can be derived as

$$\begin{aligned} \tilde{P}_b^{(b_u)}(K, \lambda, \gamma) &= \tilde{P}_e^{(X_u)}(K, \gamma) \left[\frac{MK - 1}{2M(K - 1)} - \frac{Q(\sqrt{2(1 - \lambda)M\gamma})}{M} \right] \\ &+ \frac{Q(\sqrt{2(1 - \lambda)M\gamma})}{M}. \end{aligned} \quad (68)$$

Considering a practical scenario when $\gamma \in [0 \text{ dB}, 8 \text{ dB}]$, $K \geq 512$ and $10\% < \lambda < 50\%$, $\eta(\lambda, K, \gamma)$ is the substitution for the fractional equation under the root sign in (64) and (67) and denoted as

$$\eta(\lambda, K, \gamma) = \frac{2\lambda M}{K} \gamma. \quad (69)$$

Then, $\eta(\lambda, K, \gamma)$ satisfies

$$\eta(\lambda, K, \gamma) \leq \eta(50\%, 512, 10^{0.8}) = 0.0123 \ll 1. \quad (70)$$

Furthermore, by ignoring the effect of $\eta(\lambda, K, \gamma)$ and comparing (47)–(51) with (65)–(68), we can get

$$\tilde{P}_b^{(b_u)}(K, \lambda, \gamma) \approx P_b^{(b_u)}(K, (1 - \lambda)\gamma) \quad (71)$$

and

$$[B_{\text{PCSK-CFFH}}]_{\text{dB}} \approx -10 \lg(1 - \lambda). \quad (72)$$

This means that the E_b/N_0 backoff of the PCSK-CFFH system with SC is approximately the same as that of the BPSK-CFFH system within the aforementioned range of γ , K , and λ .

IV. SIMULATIONS AND DISCUSSION

In this section, the previous theoretical results are verified by the simulations. The performances of the traditional BPSK-CFFH system are used as benchmarks. The widely used extended m-sequence [28] is adopted as the base sequence $\mathbf{c}_u$.

Extended m-sequence has good autocorrelation properties with sufficient length.

Fig. 7(a) demonstrates the SER performances of the proposed PCSK-CFFH system. From the figure, it can be observed when K is larger than 512, the gap between the overall SER and the SER for X_u is negligible. In addition, the SER for Y_u is always half of X_u for different values of E_b/N_0 . By recalling the two types of errors in Section III-A, it can be derived that most of the errors of the system come from *Type II*.

Fig. 7(b) compares the overall BER performance of PCSK-CFFH and BPSK-CFFH systems. When the required BER is 10^{-3} , it can be seen that the proposed PCSK-CFFH system with $K = 512$ and 4096 achieves the desired BER at $E_b/N_0 = 3.03$ and 2.52 dB, respectively. However, for the BPSK-CFFH system under the same conditions, the required E_b/N_0 is 6.79 dB that is much higher than that of the PCSK-CFFH system. Moreover, the total number of the transmitted information bits in each PCSK-CFFH symbol is $1 + \log_2 K = 10$ ($K = 512$) and 13 ($K = 4096$), which are much higher than that of the BPSK-CFFH system as well. It can be concluded that PCSK-CFFH is a more efficient transmission scheme compared to conventional BPSK-CFFH.

Fig. 8 illustrates the BER performance of the proposed PCSK-CFFH system over AWGN channels under hostile narrow-band jamming. As discussed in Sections II and III, each symbol of the frequency hopping system traverses all K available frequencies to obtain the full frequency diversity, and the hopping pattern is randomized by the symbol. At the receiver, the SC is adopted for PCSK-CFFH and BPSK-CFFH. The experimental and the numerical curves have been shown to assess the approximation of (68). The performance of ML reception in Section II is also given for comparison. In the simulations, the jamming signals have the same constant power in each jammed hop. Specifically, in our simulations, jamming-to-signal-ratio (JSR) of the jammed hop equals 50 dB. Additionally, the overall JSR is $\text{JSR}_{\text{overall}} = 50 \text{ dB} - 10 \lg(1/\lambda)$. The corresponding BER results with $\lambda = (1/8), (1/4), (1/2)$ are depicted in Fig. 8, respectively.

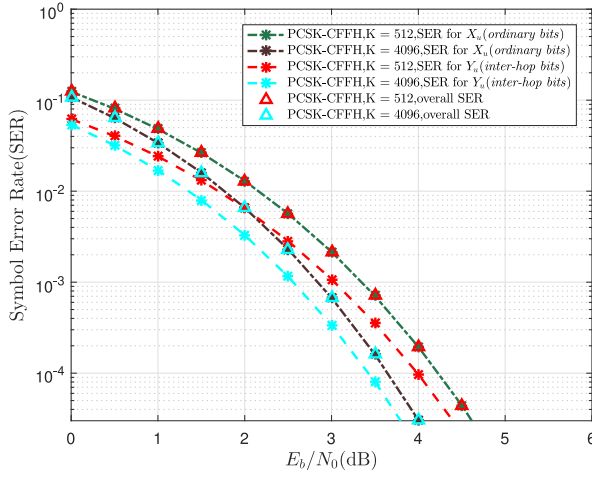
$$\begin{aligned} \tilde{P}_e^{(b_u)}(K, \lambda, \gamma) &= 1 - \frac{1}{\sqrt{2\pi}} \int_0^\infty \prod_{\substack{\tilde{X}_u=1, \\ \tilde{X}_u \neq X_u}}^K P(|x'| < v') e^{-\frac{(v' - (1-\lambda)\sqrt{E_s R_s})^2}{(1-\lambda)N_0 R_s}} dv' \\ &= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{2(1-\lambda)M\gamma}}^\infty \left[1 - 2Q\left(\frac{v + \sqrt{2(1-\lambda)M\gamma}}{\sqrt{1 + \frac{2\lambda M}{K}\gamma}}\right) \right]^{K-1} e^{-\frac{v^2}{2}} dv \end{aligned} \quad (64)$$

$$\tilde{P}_b^{(X_u)}(K, \lambda, \gamma) = \frac{K}{2(K-1)} \tilde{P}_e^{(X_u)}(K, \lambda, \gamma) \quad (65)$$

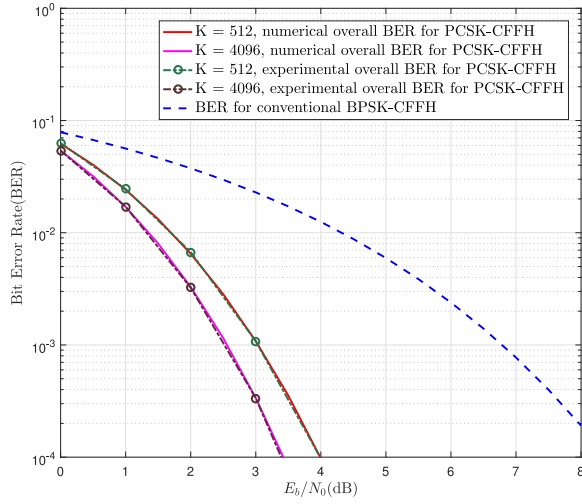
$$\tilde{P}_b^{(Y_u)}(K, \lambda, \gamma) = Q(\sqrt{2(1-\lambda)M\gamma})(1 - \tilde{P}_e^{(X_u)}(K, \lambda, \gamma)) + \frac{1}{2} \tilde{P}_e^{(X_u)}(K, \lambda, \gamma) \quad (66)$$

where

$$\tilde{P}_e^{(X_u)}(K, \lambda, \gamma) = 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \left[1 - 2Q\left(\frac{v + \sqrt{2(1-\lambda)M\gamma}}{\sqrt{1 + \frac{2\lambda M}{K}\gamma}}\right) \right]^{K-1} e^{-\frac{v^2}{2}} dv \quad (67)$$



(a)

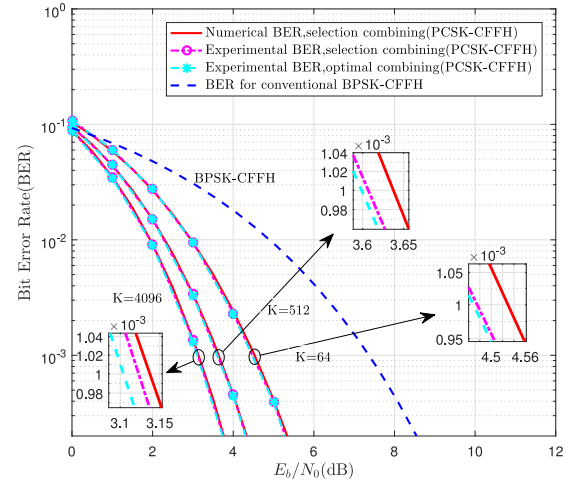


(b)

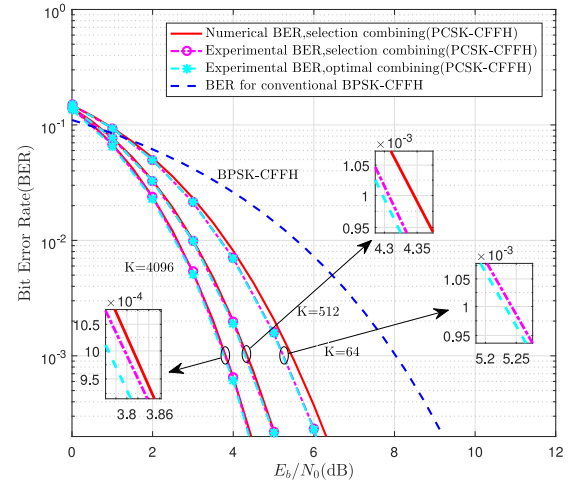
Fig. 7. Performance of PCSK-CFFH. (a) SER for PCSK-CFFH. (b) BER for PCSK-CFFH.

Comparing Fig. 8(a)–(c), it is worth noting that the experimental curves coincide with the theoretical results perfectly with a gap of less than 0.15 dB at $\text{BER} = 10^{-3}$ when $K = 512$ and 4096. This is because the Gaussian approximation in (61) works well when K is large enough. From Fig. 8(a)–(c), it can also be seen that the experimental results of SC almost overlap with those of the optimal combining with a gap less than 0.1 dB. This fact implies that for severe narrow-band jamming, the BER performance of the PCSK-CFFH system with SC is close to that with the optimal combining given in Section II. Moreover, as shown in Fig. 8(a)–(c), the SNR backoff occurs in both BPSK-CFFH and PCSK-CFFH systems. Even with the SNR backoff in the interval $\lambda = (1/8)$ to $(1/2)$, the PCSK-CFFH system is still significantly superior to the BPSK-CFFH system when E_b/N_0 is greater than 4 dB and BER less than 10^{-3} with $K \leq 512$.

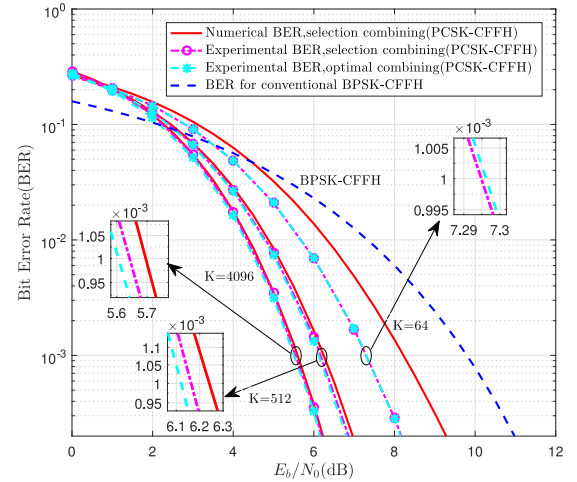
For the backoff of the PCSK-CFFH system, more detailed simulation results are shown in Fig. 9. The numerical BER curves of the PCSK-CFFH system for different λ are depicted in Fig. 9(a) and (b) in the cases of $K = 512$ and $K = 4096$,



(a)



(b)



(c)

Fig. 8. Performance of the PCSK-CFFH system in jamming scenarios. (a) $\lambda = (1/8)$, $\text{JSR}_{\text{overall}} = 41$ dB. (b) $\lambda = (1/4)$, $\text{JSR}_{\text{overall}} = 44$ dB. (c) $\lambda = (1/2)$, $\text{JSR}_{\text{overall}} = 47$ dB.

respectively. The vertical dashed lines in both figures mark the E_b/N_0 for the different conditions when the BER reaches 10^{-3} . The backoff values for $K = 512$ and $K = 4096$ with $\lambda =$

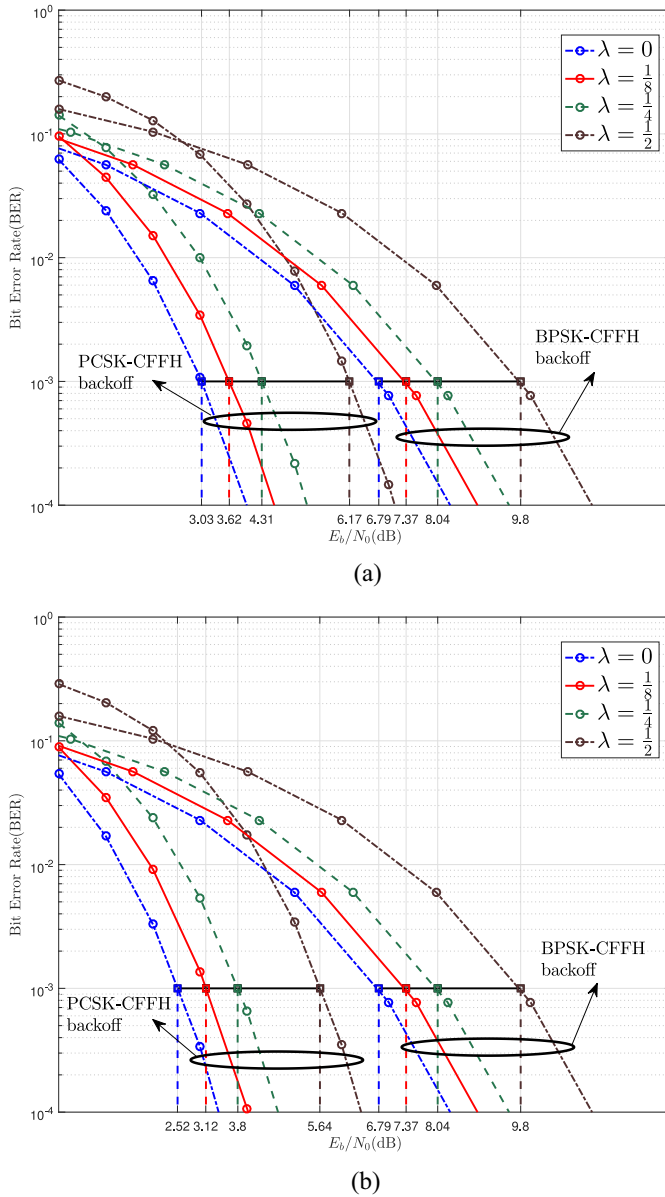


Fig. 9. Backoff performance of PCSK-CFFH compared to BPSK-CFFH. (a) $K = 512$. (b) $K = 4096$.

(1/8), (1/4), (1/2) are 0.59, 1.28, 1.28 and 1.28, 1.28, 1.28, corresponding to (72). The deterioration of backoff between the PCSK-CFFH and BPSK-CFFH is less than 0.2 dB. It can be concluded that PCSK-CFFH still has good performance when narrow-band jamming is present and the SC scheme is adopted. The BER backoff of the PCSK-CFFH system is comparable to that of the BPSK-CFFH system when $K \geq 512$.

V. CONCLUSION

In this article, we propose a new frequency hopping scheme named PCSK-CFFH which is applicable to IIoT. The basic idea of the PCSK-CFFH scheme is to conduct CCSK on the phases of interhop in order to achieve biorthogonal modulation. Compared to the conventional CFFH, the proposed scheme introduces the extra dimensions of interhop to transmit information without extra cost on either bandwidth or

power. The specific structures of the transmitter and receiver are designed for the PCSK-CFFH system in this article. These structures imply that the proposed PCSK-CFFH has a great potential to be applied to low-power and highly reliable IIoT communications. For the proposed PCSK-CFFH system, the BER performance over AWGN channels and the performance in the presence of narrow-band jamming with SC are analyzed theoretically. At the end of this article, the simulation results show the noticeable performance advantages of the proposed PCSK-CFFH system compared to the conventional BPSK-CFFH system in terms of power efficiency and spectrum efficiency.

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